

Guardian — LED Feedback Sandbox

Load your CSVs, simulate reps from real data, and watch the LED respond to good vs bad form.

[capture_veronika](#) / [simulation-dashboard-veronika.html](#)

STEP 1 — LOAD YOUR CSV FILES

Click to select all your CSV files

Select all files from capture_veronika_first_round/ and capture_veronika_5_14_26/ at once.
Dashboard auto-sorts into Good and Bad pools by filename.

Good: 12 files

Bad: 12 files

STEP 2 — RUN SIMULATION

Simulate 1 rep

Run 10-rep session

Reset

Each rep picks a random CSV from your pool (good or bad, 50/50).
Streams real accel_x data, detects peak, fires LED feedback.

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SESSION STATS

10
TOTAL REPS

4
GOOD REPS

6
BAD REPS

Bad_form1_first_round

LAST FILE USED

LED SIMULATOR — DO (YELLOW)

GPIO DO

Waiting for next rep...

Rep #10 BAD 2 blinks [Bad_form1_first_round]

Rep #9 BAD 2 blinks [Bad_form3Left]

Rep #8 GOOD 1 blink [Good_form1Left]

Rep #7 BAD 2 blinks [Bad_form3Left_first_round]

Rep #6 BAD 2 blinks [Bad_form2Left]

Rep #5 GOOD 1 blink [Good_form3_first_round]

Rep #4 GOOD 1 blink [Good_form1Left]

Rep #3 BAD 2 blinks [Bad_form2Left_first_round]

Rep #2 BAD 2 blinks [Bad_form2]

Rep #1 GOOD 1 blink [Good_form3Left_first_round]

ACCEL X — REP PLAYBACK (REAL DATA)

accel_x (m/s²)

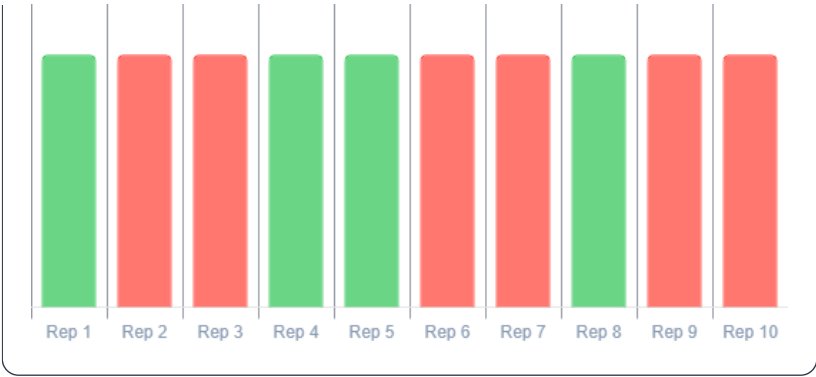
61 samples | peak accel_x = 14.667 m/s2

FORM RESULT — ALL REPS

BLINK TIMING — TUNE HERE, COPY TO ARDUINO

file:///C:/Users/pon45/Documents/515/Group Project/Guardian/capture_veronika/simulation-dashboard-veronika.html

1/2



Blink ON (ms) 200 ms

Blink OFF gap (ms) 150 ms

Post-rep pause (ms) 800 ms

Playback speed 5x

Copy these values into `led_feedback.ino`:

```
const int BLINK_ON    = 200; // ms
const int BLINK_OFF   = 150; // ms between double-blink
const int POST_PAUSE  = 800; // ms after pattern ends
```